RESEARCH STUDY COMPANION

Polynomial Image Compression

From unit distances and the sum-product phenomenon to four fixed-polynomial tracks

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All four targets remain OPEN.

Shared mechanisms, worked calculations, scoped failures, and the precise missing steps.

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This is a typeset export of the seven study notes. Their source and preservation statements describe the original Markdown preparation. The mathematical content and claim statuses are preserved; no new research or proof audit is included.

READING GUIDE

Study notes: from unit distances and sum-product to four polynomial tracks

Prepared **14 September 2026**, explaining the recorded research through **10 September 2026** and the **11 September reconciliation**.

All four fixed targets remain **OPEN**. The offline work produced supporting theorems, exact finite constructions, transfer implications, and precise reasons that several proposed methods fail. These notes explain that progress without adding a research result or changing a track's status.

This task used local sources only. It performed no proof search, mechanism pivot, expanded experiment, imported-program execution, literature search, commit, or publication. Temporary text extractions and preservation hashes are under ignored `scratch/study-notes-2026-09-14/`.

Reading order

Start with the shared explanation, then read the tracks independently.

Note	Main question it explains
Shared foundations	What can be carried over from the supplied OpenAI unit-distance manuscript and the frozen sum-product reconstruction?
T01: cubic difference plus sum	Why did the quadratic model succeed, while the tested cubic box, digital, and rich-fiber methods failed?
T02: quartic hybrid and local precision	Why are the exact C2-C4 local improvements real progress, yet insufficient globally?
T03: Eisenstein norm and adaptive radicals	How did engineered norm pairs lead to Cartesianization and then to the current signed-root problem?
T04: cubic slices and affine growth	Why do rich fibers and growing affine inputs still require a bound on the entire output?
Sources and preservation record	Which versions support the notes, what was inspected, and what remains unavailable?

The intended background is elementary algebra and finite-set counting. The shared note introduces the number-field language used later. Each track has a worked calculation, a precise unresolved endpoint, and recall questions whose answers appear in the note. The questions are study exercises, not new research assignments.

Four separate endpoints

Track	Fixed polynomial	Recorded progress	Missing conclusion
T01	$(x-y)^3+x+y$	Scoped continuous/digital obstructions and an exactly injective tested rich-fiber family	No supported cubic construction remains. The separate x^2+xy+y^2+x transfer has no witness.
T02	$(x+2y)^4+3x-y$	Analytic local counts through C4, including saturation at $22/49$ after both input costs	A concrete subcarrier and full-support bound with a positive net rate
T03	x^2+xy+y^2	Exact adaptive inputs and finite image counts through $(32, 269)$; root-cover identities and scoped failures	Uniform RC15/4 on the actual adaptive levels
T04	$(x-y)^3-3(x+y)$	Proved distinct-input entropy for one complementary-unit affine system	A strict full-image exponent gap for that system

This table restates the controlling state records S-T01-S-T04. It does not transfer results between tracks. The shared corrected rank-one quadratic theorem solves none of these four fixed targets.

How to read the claims

Each numbered study item has an ID, a status, explicit scope, and a source locator. These are **documentary claim records**, version 2026-09-14; they are not newly proved mathematical claims. The polynomial, quantifiers, hypotheses, dependencies, and remaining gap are stated in the item or its note's opening. Source versions and hashes are in SOURCES.md.

- **PROVED, reported** preserves an identified proof status in the source system. Reading its statement does not independently verify its proof.
- **PROVED, argument inspected** means the relevant written argument was read for explanation here. No independent proof audit or dependency recertification was performed.
- **CONDITIONAL** marks an application with an unproved premise, even when the implication itself is proved.
- **FAILED** applies only to the named family, inference, or counting method.
- **EXPERIMENTAL** marks recorded finite evidence, with its range and arithmetic specified; no mathematical computations were rerun for these notes.
- **EXTERNAL RESULT** identifies supplied external material. The sum-product result is accessed through the frozen reconstruction, not a newly retrieved primary paper. No publication-status or novelty check was performed.
- **HISTORICAL / SUPERSEDED** preserves earlier reasoning while identifying the later controlling correction. Explanatory interpretations add no theorem.

Preservation boundary

Only this new dated study folder is intended as a repository addition. Existing SPECs, STATUS files, archives, manifests, copied references, research code, and inherited uncommitted changes are preserved. The notes are a reading companion; they do not become a replacement checkpoint or a new task queue.

STUDY NOTE 1

Shared foundations: what the source constructions contribute

Study index · Source identities

Version: 2026-09-14; research cutoff: 2026-09-10. This is an explanation of supplied results, with no independent proof audit. In this note, $\mathcal{P}$ denotes an additive carrier, leaving P for a polynomial.

SN-S1 - Two achievements with different counting goals

EXTERNAL RESULT, supplied statement and relevant arguments inspected. The supplied OpenAI manuscript *Planar Point Sets with Many Unit Distances* states that a fixed $\delta > 0$ and infinitely many n admit an n-point planar set with at least $n^{1+\delta}$ unordered unit-distance pairs. Its proof counts selected pairs whose difference is a norm-one translation. It does not claim that every pair has one of a small number of distances. Source: **S-UD**, Theorem 1.1, Proposition 2.2, Theorem 2.3 and Lemmas 2.4-2.6.

EXTERNAL RESULT, reported through the inspected frozen reconstruction. Phase A attributes to Bloom-Sawin-Schildkraut-Zhelezov a construction of arbitrarily large finite real sets with, for fixed $c > 0$,

$$\max\{|A + A|, |AA|\} \leq |A|^{2-c}.$$

Here the entire sumset and product set are bounded. The primary paper is not in the recovered bundle used here; its bibliographic identity is recorded as given by Phase A, without new retrieval. Sources: **S-A**, §§1, 11-13; **S-A-LOG**.

Interpretation. The polynomial project asks for another full-support statement: all $|A|^2$ ordered pairs must land in at most $|A|^{2-c_P}$ values of one fixed P. Neither many selected edges nor separately small sum and product sets automatically supplies the required mixed-polynomial bound.

SN-S2 - Why number fields help

EXTERNAL RESULT / explanatory unpacking of S-UD and S-A. A totally real number field K of degree d has d real embeddings σ_i. Its algebraic integers become a lattice in $\mathbb{R}^d$ under $\alpha \mapsto (\sigma_1(\alpha), \ldots, \sigma_d(\alpha))$. The discriminant Δ_K measures its density: the lattice covolume is $\sqrt{\Delta_K}$. A bounded root discriminant means $\Delta_K \leq D_0^d$ for a fixed D_0, so this density loss is exponential in degree and can be compared with an exponential gain.

For a nonzero algebraic integer z, the product of its conjugates is a nonzero integer. Thus not all $|\sigma_i(z)|$ can be below one. This gives separation for lattice packing. One real embedding is injective and preserves polynomial evaluation exactly; projection to $\mathbb{R}$ loses no cardinality. Source: **S-A**, §§3-4; **S-Co**, Lemmas 5.2-5.3 and §6.

EXTERNAL RESULT, supplied arithmetic theorem inspected. The unit-distance paper supplies totally real unramified tower layers F_j of growing degree, constant root discriminant, and fixed rational primes $q_b \equiv 1 \pmod 4$ splitting completely in every layer. Its construction first fixes a sufficiently large ℓ, giving $t = \lfloor (\ell - 1)^2/100 \rfloor$ such primes and $\log \mathrm{rd}(F_j) = O(\ell \log \ell)$, then lets the layer grow. The associated CM fields $F_j(i)$ are complex and have a different role. Source: **S-UD**, Proposition 3.8(P1)-(P6), Remark 3.9.

The paper's norm-one translations arise from binary choices of conjugate prime ideals. Selecting a common ideal class converts enough choices into actual elements, with a class-number loss and fixed denominator. Their absolute values are one at every complex embedding. Window averaging then retains many translation edges. Source: **S-UD**, Proposition 2.2 and §2.1.

SN-S3 - The sum-product mechanism, step by step

EXTERNAL RESULT, reconstruction inspected; no new primary-source audit. In a totally real field, define

$$B^+(T) = \{b \in \mathcal{O}_K : |\sigma_i(b)| \le T \text{ for all } i\},$$

$$\mathcal{P} = X + B^+(\varepsilon X), \qquad G = \{g \in \mathcal{O}_K^\times : |\log|\sigma_i(g)|| \le Y \text{ for all } i\}.$$

Choose a fixed sufficiently small $\varepsilon > 0$ and the source's sufficiently large fixed Y, X, with X integral. The construction is $A = G\mathcal{P}$.

The additive carrier occupies a short positive window around X in every embedding. If $g_1p_1 = g_2p_2$, then the unit $g_1/g_2 = p_2/p_1$ is uniformly close to one in all embeddings. The source's unit-separation lemma forces it to be one. Consequently $|A| = |G||\mathcal{P}|$.

Logarithms turn unit multiplication into addition. Hence GG lies in a log box of radius $2Y$, and the full product count satisfies

$$AA = (GG)(\mathcal{P}\mathcal{P}), \qquad \frac{|AA|}{|A|^2} \le \frac{|GG|}{|G|^2}.$$

The same units have $|\sigma_i(g)| \le e^Y$. Every sum in $A + A$ therefore lies in one ordinary box of radius $4Xe^Y$. These two different mechanisms control the two full supports.

The frozen report gives, for disposable absolute $c_0 > 0$,

$$\frac{(c_0XY)^d}{Y\Delta_K^{3/2}} \le |A| \le (XY/c_0)^d,$$

$$\frac{|AA|}{|A|^2} \le c_0^{-d}Y^{1-d}\Delta_K^2, \qquad |A + A| \le (e^Y/c_0)^d\Delta_K^{1/2}|A|.$$

Fix D_0, choose Y to buy product saving, choose X to pay the additive distortion, then grow d. The regulator controls the density of the unit lattice and is paid through the report's bound $R_K \le \Delta_K$ for totally real $d \ge 2$.

Sources: **S-A**, §§3-8, 10-12; **S-A-LOG**, corrections 1-8.

Interpretation. The final sum-product construction uses the unit lattice directly. It does not reuse the unit-distance proof's split-prime, ideal-class, fixed-denominator construction. The shared feature is a favorable comparison of exponential gains and costs, not an identity of the two constructions.

SN-S4 - What Phase B extracted

PROVED, reported abstraction. Write $r_P(z) = |\{(a,b) \in A^2 : P(a,b) = z\}|$ and $E_P(A) = \sum_z r_P(z)^2$. Cauchy-Schwarz gives

$$|P(A,A)| \ge |A|^4/E_P(A).$$

A large energy lowers this lower bound; it does not upper-bound the image. Phase B therefore requires a complete output container, a factorization through a small support, or coverage of almost every pair with a charged tail. Source: **S-B**, Validated abstraction principles 1-3; **S-MASTER**, §5.

PROVED, reported local-to-global rule. Complete splitting gives $\mathcal{O}_K/p\mathcal{O}_K \cong \mathbb{F}_p^d$. A scalar residue restriction of density q_p becomes a restriction of density q_p^d. To turn that into a global upper bound, count permitted classes and multiply by a **uniform worst-case bound within each class**. Multiplying density by a total box count without this step is unjustified.

If input rates are $e^{ad-o(d)} \le |A_d| \le e^{Bd+o(d)}$ and the full output rate is at most $e^{(c-s)d+o(d)}$, the required margin is

$$\delta = 2a - c + s > 0.$$

Every fixed exponent below δ/B is then certified after absorbing errors. An input loss of $e^{-\eta d}$ costs 2η in this comparison because both inputs are restricted. Source: **S-B**, Validated abstraction principles 4-6; **S-MASTER**, §6; **S-REG**, mechanism library. This is a gate for a proved construction, not evidence that its premises hold.

SN-S5 - The shared positive result: corrected C0

PROVED within the supplied source system; argument inspected. For fixed integral a, b, c, e with $\gcd(a, b) = 1$ and $ab(ae - bc) \neq 0$,

$$P(x, y) = (ax + by)^2 + cx + ey$$

admits arbitrarily large same-input real sets with a fixed positive saving depending on P. Source: **S-Co**, Theorem Co.1, §§5-6; **S-Co-LOG**.

The key calculation is short enough to retain. Let $L = ax + by$, $M = cx + ey$ and $P_Q = L^2 + QM$, so $P_Q(Qx, Qy) = Q^2 P(x, y)$. At selected odd primes dividing fixed Q, outputs are squares. Put

$$\rho = \prod_{p \mid Q} \frac{p+1}{2p}, \quad C_P = 2((|a| + |b|)^2 + |c| + |e|) + 1.$$

The source fixes its tower and primes with $\theta = C_P D \rho < 1$, then fixes $X > \max(Q, \sqrt{D})$. Its centered carrier $B_j \subset \mathcal{O}_{F_j}$ obeys

$$(X/\sqrt{D})^{d_j} \leq |B_j| \leq (2X + 1)^{d_j}, \qquad |P_Q(B_j, B_j)| \leq \theta^{d_j} |B_j|^2.$$

Taking $A_j = Q^{-1} \sigma_j(B_j)$ gives the fixed target, with exponent $-\log \theta / \log(2X + 1)$. This calculation uses totally real F_j, charges all classes uniformly, and covers every pair. The source has explicit bounds here; the general asymptotic gate above requires strict exponent loss.

Scope. Rank two, degree three, and degree four are separate obligations. This theorem includes $(x + 2y)^2 + 3x - y$, not T02's quartic. No novelty claim is made: the master handoff retains a literature-priority caution.

Recall check

1. What replaces the unit-distance edge count in this project? **A bound on the complete polynomial image, including every ordered pair.**
2. Why prove both input bounds? **The lower bound proves growth and pays costs; the upper bound converts a degree saving into a power of input size.**
3. Why are norms useful without solving the problem? **They supply exact identities and separation, but a common Cartesian carrier and a small entire output support still have to be proved.**

STUDY NOTE 2

T01 study notes: cubic difference plus sum

Study index · Shared foundations · Source identities

Version: 2026-09-14; research cutoff: 2026-09-10. The fixed target is **OPEN**:

$$P_+(x,y) = (x-y)^3 + x + y, \qquad \exists c > 0\ \forall N_0\ \exists A \subset \mathbb{R}\ \text{finite} : |A| \geq N_0, \quad |P_+(A,A)| \leq |A|^{2-c}.$$

The progress below is source-reported unless explicitly marked as an inspected argument. No new proof audit or experiment was performed. Source **S-T01**, §§1-2, fixes this identity; **S-T01-FINAL** concerns a separate quadratic branch.

SN-T01-1 - The successful model and the cubic difficulty

PROVED, reported. The rank-one benchmark $F_2 = (x-y)^2 + x + y$ has a same-set power-saving construction. In the supplied tower, scaling algebraic-integer inputs by a fixed denominator M makes the cleared output

$$M^2 F_2(\alpha/M, \beta/M) = (\alpha-\beta)^2 + M(\alpha+\beta).$$

Modulo each selected split prime dividing M, this is a square. Repetition over the field embeddings supplies many residue restrictions, while uniform packing counts every allowed output. Source: **S-T01**, §4.1 and (T01.3); the quadratic checkpoint locator is given there.

FAILED, specified cubic certificate. For the cubic, clearing denominators gives

$$Z_N(U,V) = (U-V)^3 + N^2(U+V), \qquad N^3 P_+(U/N, V/N) = Z_N(U,V).$$

Cube residues can be sparse, but the denominator and cubic output width cost more. For the source's one-prime certificate, $q \equiv 1 \pmod{12}$, $q \geq 13$, the normalized bound has infimum $R\delta_{q,2}(2q+1) > 8R$, where R is the root discriminant and $\delta_{q,2} = (q^2-q+3)/(3q^2)$. A base greater than one cannot prove an exponent saving. This is a failed upper-bound method, not a lower bound on all sparse cubic images. Source: **S-T01**, §6.1.

Interpretation. The source tower survives the adaptation; the favorable counting balance does not. “More split coordinates” amplifies the loss too.

SN-T01-2 - Why the literal sum-product transfer stalled

FAILED, reported parameter-count method. Substitute two independent elements gp, hq of a Phase-A carrier into the cubic:

$$P_+(gp, hq) = p^3 g^3 - 3p^2 q g^2 h + 3pq^2 g h^2 - q^3 h^3 + pg + qh.$$

For products alone, gh is a useful compressed label. Here the joint labels (g^2h, gh^2) recover (g,h) over the totally real field. Restricting to one common scale controls only a $1/|G|$ fraction of pairs. Counting these mixed labels therefore supplies no full-square saving; it does not prove that every possible scalar sum of the terms is distinct. Source: **S-T01**, §6.8.

The unit-distance translation route has another clear limitation. For a fixed difference d,

$$P_+(a, a-d) = d^3 + 2a - d$$

is injective in the anchor a. Many selected edges of difference d give many outputs along that slice. Cross-slice collisions would need a separate theorem. Source: **S-T01**, §§4.5, 6.7.

SN-T01-3 - What the negative results actually establish

PROVED, reported scoped obstructions. The cubic work replaced several unsupported hopes with precise boundaries:

Attempt	Recorded result	Scope retained
Positive-volume blocks	For bounded Borel $W \subset \mathbb{R}^m$ of volume $V > 0$, the continuous cubic image needs more than $(\sqrt{3}/\pi)^m V^2$ unit cubes	Rules out the stated volume/covolume plus continuous-cover certificate; does not count arbitrary sparse lattice images
Regular digital sets	For fixed base, finite strongly connected right-resolving input graph, fixed start, all states terminal, and growth $\mu > 1$, $\lvert Z_{b^n}(D_n, D_n)\rvert \geq c_G \lvert D_n\rvert^2$ eventually	Excludes the proposed all-output automaton with $\lambda^3 < \mu^2$
Fixed-prime positive-entropy digits	Fix finite nonempty prime support S and $\alpha > 0$. For large S-smooth N and $D \subset [0, N) \cap \mathbb{Z}$, $\lvert D\rvert \geq N^\alpha$ implies $\lvert Z_N(D, D)\rvert \geq \eta(S, \alpha)\lvert D\rvert^2$	Includes nonregular rules under these hypotheses; does not cover growing prime support or zero entropy
One quadratic rich-fiber family	The precisely defined $A_j = \sigma_1(T_j/(2\sqrt{\zeta_j}))$ of §4.4 has exactly $\lvert A_j\rvert^2$ cubic outputs	An actual image theorem for that family, unlike the failed box certificates

Sources: **S-T01**, §§4.2-4.4, 6; **S-REG**, MC-01 and T01 failure entries. For the last row, retain the source's largest-fiber construction, translated box, $\rho_2 R \leq 1/100$, $M \geq 64$, and $X = 2\lfloor M/8\rfloor \geq 4\sqrt{R}$; it is not a theorem about arbitrary quadratic fibers.

HISTORICAL / SUPERSEDED. The old SPEC still describes regular digital witness existence as open. The later obstruction controls the route's status; the SPEC remains preserved. No narrower supported positive cubic candidate survives in the frozen record. Sources: **S-T01**, §§13-15; **S-REG**, MC-01.

SN-T01-4 - Separate norm-trace progress, with its missing witness

PROVED, argument inspected; CONDITIONAL application. The 10 September checkpoint separately studied $P_0 = x^2 + xy + y^2 + x$. Completing its norm gives

$$P_0(x, y) = q(x + 2/3, y - 1/3) - 1/3, \qquad q = x^2 + xy + y^2.$$

The shifts differ, so a homogeneous same-set witness does not automatically work. The checkpoint proves a precise repair. Suppose growing real C, $N = \lvert C\rvert$, satisfy both

$$\lvert q(C, C)\rvert \leq K_* N^{2-c_0}, \quad \lvert C - C\rvert \leq D_* N^{1+\eta}, \quad 0 < c_0 \leq 1, \quad 0 \leq \eta < c_0/2,$$

with fixed constants. Choose a richest nonzero difference δ, put $B = C \cap (C - \delta)$ and $A = B/\delta + 1/3$. Then

$$\lvert A\rvert \geq \frac{N^{1-\eta}}{2D_*}, \qquad P_0(A, A) \subseteq \delta^{-2} q(C, C) - 1/3.$$

The resulting exponent before constant absorption is

$$c_1 = \frac{c_0 - 2\eta}{1 - \eta} > 0.$$

The 2η in the numerator pays the thinning in both coordinates. This is a proved transfer implication with **no supplied witness satisfying both premises**. It has no established cubic consequence. Source: **S-T01-FINAL**, §§2.1-2.3; **S-T01**, §12.

PROVED bound, argument inspected / FAILED power-saving certificate. The tested shifted Phase-A family $A_j = \sigma_1(G_j\mathcal{P}_j) - 1/6$ has an all-pair reflection bound $(N_j^2 + N_j)/2$. Its full box estimate has output rate $g > 2h$, where h is the proved input lower rate. Reflection saves a constant factor, not a fixed power. The actual family is not disproved. Source: **S-T01-FINAL**, §5, especially (5.4)-(5.6).

OPEN. More general solvable extensions supply no demonstrated additional output gain. The checkpoint completed its bounded study and adopted no new amplification conjecture. Its arithmetic literature survey predates the freeze; these notes do not repeat it. Source: **S-T01-FINAL**, §§4, 8.

Recall check

1. Why does a quadratic theorem not survive replacing the square by a cube? **The full output width and denominator costs change; the residue saving must be compared with them again.**
2. Which negative result above determines an actual image exactly? **The specified rich-fiber family. A failed enclosing bound alone does not.**
3. What is missing from the separate quadratic transfer? **A growing C with both small $q(C, C)$ and sufficiently small $C - C$. The implication is proved; its application remains conditional.**

STUDY NOTE 3

T02 - the quartic and the difference between local savings and global compression

Prepared 2026-09-14 from the record frozen through 2026-09-10. Target:

$$P_4(x,y) = (x+2y)^4 + 3x - y.$$

OPEN. No recorded construction establishes a fixed power saving for the entire image of arbitrarily large same-set inputs. This note explains existing arguments; it does not extend their scope or rerun their computations.

What the two papers contributed

The shared work supplied number-field carriers, bounded real projections, direct products of units and lattice points, and a quantitative way to charge input density against output density. T02 tested whether a special local quartic image could make those carriers compress. The central progress was increasingly precise local counting, followed by a precise identification of why a specified global packing argument cannot use that saving successfully. See shared foundations, **S-A**, and **S-T02**.

SN-T02-1 - the hybrid carrier and the determinant-seven relation

PROVED, reported; definitions and counting explanation inspected. Track T02, fixed quartic above. Sources: **S-T02**, §§2-4; **S-T02-FINAL**; **S-C2-LOG**. These are existing conclusions, not a new proof audit.

Earlier work considered the deformation $F_Q(x,y) = (x+2y)^4 + Q^3(3x-y)$, for which $F_Q(Qx,Qy) = Q^4 P_4(x,y)$. The later C4 carrier uses $Q = 1$. This scaling identity is necessary when moving between the deformation and the fixed target; a theorem about variable coefficients alone would not suffice.

The C2 hybrid uses positive units and a translated box with parameters chosen to make the representation of each input unique. For a ratio $u = g_1/g_2$ and scale $v = g_2$, a pair of box elements contributes

$$K_u = \{((ut_1+2t_2)^4,\ 3ut_1 - t_2) : t_1, t_2 \in T\}, \qquad v^4\alpha + Q^3 v\beta.$$

The recorded positivity hypotheses give $|K_u| = |T|^2$. Also $\sum_u |G(u)| = |G|^2$. Thus simply counting these parameters gives the original $|A|^2$ pair count. Compression requires overlaps between their scalar outputs. This parameter calculation is **not** an actual-image lower bound.

The useful local relation appears under the linear change

$$(t_1,t_2) \longmapsto (a,b) = (ut_1+2t_2,\ 3ut_1 - t_2).$$

For integral unit u, its determinant is $-7u$ and its image satisfies $a + 2b \in 7\mathcal{O}$. The index is 7^d in degree d. The determinant relation and its scalar residue consequence describe the same restriction: they cannot be charged as two independent savings.

SN-T02-2 - what C3 and C4 actually prove locally

PROVED, reported; C4 local argument inspected for exposition. Sources: **S-C3-LOG**, **S-C4**, §§2-5, and **S-T02-FINAL**.

At the first ramified level, the scalar map is $h_s(z) = 2z^4 + sz$, $s \in \{1, -1\}$, over $R = \mathbb{F}_7[\pi]/(\pi^3)$. Four regular residue classes contribute 49 images each and three critical classes contribute four each: **208** images in total. C3 proves analytically, for every nonempty $E \subset R$, the bound

$$\frac{343\,|h_s(3E - E)|}{|E|^2} \geq \frac{4}{7}.$$

Equality occurs for a full critical fiber $E = a + \pi R$, $a^3 = -s$. The archived structured enumeration supports the investigation; it was not an enumeration of all 2^{343} subsets and is not the proof of this universal claim.

C4 follows a full critical fiber to higher precision in an actual local ring: $7\mathcal{O} = \mathfrak{p}^3$, residue degree one, $R_m = \mathcal{O}/\mathfrak{p}^{3m}$, and $E_m = (\alpha + \mathfrak{p}) \bmod \mathfrak{p}^{3m}$, where α is the integer lift of one of $3, 5, 6$ and $\alpha^3 = -1 \bmod \mathfrak{p}$. Then

$$|R_m| = 7^{3m}, \quad |E_m| = 7^{3m-1}, \quad N_m := |P_4(E_m, E_m)| = \begin{cases} 4 & m = 1, \\ 22\,7^{3m-4} & m \geq 2. \end{cases}$$

Here is the mechanism behind the number 22. Set $L = x + 2y$, so

$$P_4(x, y) = g(L) + \frac{7}{2}x, \qquad g(L) = L^4 - \frac{1}{2}L.$$

The independent variables range through $(3\alpha + \mathfrak{p}) \times (\alpha + \mathfrak{p})$. The second summand fills a translate of $\mathfrak{p}^4$. Expanding g about $L_0 = 3\alpha$ and writing the increment as $\pi b + \pi^2 c$, the relevant terms modulo $\mathfrak{p}^4$ are (the linear term vanishes because $g'(L_0) \in 7\mathbb{Z}_7 \subset \mathfrak{p}^3$)

$$A_0 b^2 \pi^2 + (2A_0 bc + B_0 b^3)\pi^3, \qquad A_0 = 6L_0^2, \quad B_0 = 4L_0.$$

A_0 is a unit. The case $b = 0$ gives one class; the three nonzero squares b^2 each allow seven next coefficients. Hence $1 + 3 \cdot 7 = 22$. The local image consists of three shallower cosets and one deeper coset. Lifting them yields the formula for every $m \geq 2$. At these higher levels the ring has mixed characteristic; replacing it by a longer truncated polynomial ring over $\mathbb{F}_7$ would change the problem.

Normalize the image density by the square of the input density:

$$\Psi_m = \frac{N_m/|R_m|}{(|E_m|/|R_m|)^2} = \begin{cases} 4/7 & m = 1, \\ 22/49 & m \geq 2. \end{cases}$$

There is an improvement at the second level and then saturation. Increasing precision does not repeatedly earn a fresh factor $22/49$. C4 counts this specified full fiber; it is not a theorem optimizing every subset at every higher precision.

SN-T02-3 - why the specified global certificate is retired

FAILED, for the stated full-interval packing certificate; PROVED, reported failure inequality. Sources: **S-C4**, §§6-8; **S-C4-RATE**; **S-T02-FINAL**. No actual-image lower bound follows from this failure.

For the recorded totally real tower, let D be its fixed root discriminant and let I_j be the product of primes above seven. Its norm is $7^{d/3}$. The carrier is the real projection of

$$C_j = -1 + \{u \in I_j : |\sigma(u)| \leq X \text{ for every real embedding}\}.$$

The lower input exponent is $a = \log X - \frac{1}{3}\log 7 - \frac{1}{2}\log D$, with upper count $(2X + 1)^d$. The chosen $X = 2\,7^{1/3}\sqrt{D}$ gives $a = \log 2$. Projection through a fixed real embedding is injective. Membership in a local fiber does not assert that this bounded carrier fills it.

The full archimedean output interval has width $W = 81(X + 1)^4 + \frac{3}{2}X + \frac{27}{16}$. Combining this interval with the local support gives the upper certificate

$$U_m = N_m^{d/3}(1 + W/7^m)^d.$$

For $m \geq 2$, its signed rate gap is

$$\kappa_m = \log \frac{X^2}{D(22/49)^{1/3}(W + 7^m)} < 0.$$

The quartic width overwhelms the fixed local gain. The final record also retires saturation mixtures and finite covers of the entire local product when every branch is charged by the specified full interval and uniform norm packing, within the stated equal-radius carrier scope. A lower bound on the size of this *upper certificate* does not show the actual image is large.

SN-T02-4 - the precise remaining task

OPEN. Source: **S-C4-RATE**, final obligation; **S-T02-FINAL**. One needs a new, fully charged bound for the actual carrier or an explicitly thinned subcarrier. If thinning costs $e^{-\eta d}$ in input size, the pair budget loses $2\eta d$. A sufficient certificate would bound the whole image by $\exp((2(a - \eta) - \varepsilon)d + o(d))$, with fixed $0 \leq \eta < a$ and $\varepsilon > 0$, while preserving upper and lower input counts. All branches and uncovered pairs must be included.

The record supplies no witness meeting this obligation. Local C4 counting remains progress even though its specified global application fails. The C4 acceptance record covers its named materials; it does not constitute external acceptance of every companion argument or of the target itself.

Recall check

1. Why divide output density by **squared** input density? **Restricting the same input set restricts both coordinates.**
2. Does deeper precision give unbounded savings? **No: $\Psi_m = 22/49$ for every $m \geq 2$ in this full-fiber construction.**
3. What failed globally? **A specified interval-and-congruence upper certificate, not every possible construction for the quartic.**

STUDY NOTE 4

T03 - Eisenstein norms, Cartesian completion, and adaptive radicals

Prepared 2026-09-14 from the record frozen through 2026-09-10. Target:

$$Q(x,y) = x^2 + xy + y^2.$$

OPEN. The latest frontier is an unproved support bound on the actual adaptive radical sequence. All computations discussed here are inherited; none were rerun or expanded for these notes.

What the two papers contributed

Writing $Q(a,b) = N(a - \omega b)$, where $\omega^2 + \omega + 1 = 0$, connects the target to multiplicative algebra. The unit-distance construction suggests organizing many representations of a norm. The sum-product construction suggests combining multiplicative structure with controlled additive sets. T03 exposes the difficulty of turning these ideas into a Cartesian square: many engineered pairs with few norms need not cover all pairs of their coordinate set. Sources: **S-T03**, §§2-7, and shared foundations.

SN-T03-1 - why a norm identity is only the beginning

PROVED, reported identities and scoped obstructions; OPEN target. Sources: **S-T03**, §§3-7; **S-REG**. Track T03, fixed Q above.

The quadratic form has rank two (matrix determinant $3/4$). It is not the rank-one square covered by corrected C0. A norm-one Eisenstein multiplier $r + s\omega$ acts on coordinates by

$$(a,b) \longmapsto (ra + sb,\ -sa + (r - s)b).$$

This preserves Q but generally introduces denominators or moves coordinates outside the proposed input set. The recorded common-box/stable-core strategy does not recover enough inputs after those costs. This is a failure of that strategy, not a universal obstruction to norm-one actions.

Likewise, rank-two local norm behavior does not automatically inherit the rank-one residue saving. The record distinguishes weighted tensorized inequalities from the ordinary unweighted support bound required here. A weight is useful only with a charged comparison to an actual same-set input.

SN-T03-2 - engineered norm pairs versus all coordinate pairs

PROVED, reported counts and obstructions; EXPERIMENTAL/finite exact examples as explicitly identified. Source: **S-T03**, the CR/PTCR and Cartesianization sections. No archived program or serialized object was executed.

A balanced Eisenstein word construction uses four letters at each of m positions, schematically $\{1, r_i, \pi_i, \bar{\pi}_i\}$. It supplies 4^m engineered pairs and at most 3^m engineered norms. If A is the union of their coordinates and $n = |A|$, a direct bound for the completed Cartesian image is only

$$|Q(A,A)| \le 3^m + (n^2 - 4^m).$$

The second term counts pairs not covered by the engineered construction. It must itself have a power saving. The recorded untwisted family instead satisfies $n^2 - 4^m \ge n^2/3 + 2^m$.

A parity twist by $1-\omega$, whose norm is three, preserves the useful word and norm counts. Exact Cartesian closure still fails for the specified PTCR family when $m \geq 2$. An approximate closure question survives historically, but it is not a proved substitute for the current adaptive-sequence obligation.

The recorded six-point example $A = \{-25, -10, -5, -2, -1, 26\}$ is instructive: 16 engineered ordered pairs have nine values, whereas the full 36-pair image has 19 values. These are reported finite exact counts. Repeating this pattern leaves engineered-pair density $(4/9)^j$ after j repetitions, so its attractive small example does not establish an asymptotic construction.

SN-T03-3 - the actual adaptive radical sequence

PROVED, reported algebraic identities and finite certificates; OPEN uniform recurrence. Sources: **S-To3**, §§3-4, 13-15; **S-To3-ZIP**, especially `TCRC_History_Parameters.md` and `TCRC_Root_Algebra_Pair_Maps.md`.

For finite $A \subset \mathbb{R}$, put $M = \max|A|$ and choose $C > M$. Define

$$L_C(A) = \left\{ \pm\sqrt{\frac{C+a}{C-M}} : a \in A \right\}.$$

The positive radicands are distinct, so this doubles the input size exactly. The recorded sequence starts with $A_0 = \{-1, 1\}$ and, at each level, chooses the parameter minimizing the next image among the specified collision-legal parameters together with $M+1$, with the prescribed least-parameter tie break. Statements about a different choice at one level describe a different descendant. They do not automatically establish a recurrence for this sequence.

After removing the common factor $C - M$, the full signed image is

$$T_C(A) = \{2C + a + b \pm \sqrt{(C+a)(C+b)} : a, b \in A\}, \quad Q(L_C(A), L_C(A)) = (C-M)^{-1} T_C(A).$$

To count it, work over $K = \mathbb{Q}(A, C)$ and group $C + a$ by their square classes in K^*/K^{*2}. Let their sizes be c_j, and let $W = |T_C(A) \cap K|$. The recorded exact decomposition is

$$|T_C(A)| = n^2 - \sum_j c_j^2 + W.$$

Pairs in different square classes contribute the outside-field part. Within a class, write $C + a = \gamma_j u_a^2$ with $u_a \in K$ and include both signs in U_j. The remaining support is a union of $\gamma_j Q(U_j, U_j)$. Thus a dominant square class is useful, but its own signed-root image still needs a support bound.

The inherited exact levels are:

Level m	$\lvert A_m \rvert$	$\lvert Q(A_m, A_m) \rvert$	Chosen next C
0	2	2	**5/3**
1	4	5	**5/2**
2	8	17	**4**
3	16	68	**3**
4	32	269	Not established in the retained record

For the level-three choice $C = 3$, square-class sizes are $4, 2, 2$ and eight singletons. Their squared sizes sum to 32. Therefore 224 values lie outside the base field; the inside-field support has size 45; the total is 269. This is an explanation of the recorded certificate, not an extrapolation. The recorded field degree 16,384 at level four is not a

compression exponent: here the growing input parameter is the iteration depth m.

SN-T03-4 - what the finite cover investigations established

PROVED, reported finite certificates; OPEN uniform extension. Source: **S-T03-ZIP**, all five identified TCRC reports. Their statements were read; their supporting programs and raw objects were not rerun or loaded.

One tempting approach is to cover the signed-root image by a few affine copies of the previous image. The retained small cases show why the number and shape of the copies matter. At level one, $C = 17/8$ produces root-image size 12 against old image size five, but requires four affine copies in the reported cover. At level three, $C = 3$ has root-image size 17 and requires five affine copies. These examples defeat particular small-cover expectations; they do not disprove every eventual bounded-cover assertion.

Another level-three choice, $C = 4$, has a six-element shifted square-class core with ten exceptions. Its signed-root image has size 41 and fits the stated one-copy image cover. But the simple exceptional-pair charge is $10(2 \cdot 16 - 10 + 1) = 230$. The resulting bound is $41 + 230 = 271$, already larger than the actual optimizer's 269. A good core alone has not paid for all pairs.

Repeating $C = 4$ at the next actual input gives at least 929 values, exceeding $3 \cdot 269 = 807$. This rejects that repeated parameter as the proposed factor-three mechanism; it does not determine the next optimizer.

SN-T03-5 - the missing RC15/4 statement and its conditional payoff

OPEN premise; CONDITIONAL target conclusion via the reported implication. Source: **S-T03**, final adaptive-level obligation and rate ledger.

The required statement concerns every sufficiently large **actual** level. It asks for fixed $H < \infty$ and m_0 such that, for every $m \geq m_0$, there are $C^\dagger \in F_m = \mathbb{Q}(A_m)$ with $C^\dagger > M$, a subset $B \subset A_m$, and a common positive square-class representative $\gamma \in F_m$ satisfying

$$|A_m \setminus B| \leq H\sqrt{n}, \qquad C^\dagger + a = \gamma u_a^2 \quad (u_a \in F_m,\ a \in B),$$

and, with $U = \{\pm u_a : a \in B\}$ and $s_m = |Q(A_m, A_m)|$,

$$|\gamma Q(U,U)| \leq \frac{15}{4} s_m + Hn^{3/2}.$$

The field condition is part of the obligation. It cannot be met by simply adjoining every desired square root and then ignoring the extension cost. One must also justify the comparison with the defined optimizer.

Writing $e = |A_m \setminus B|$, the exceptional signed-pair charge is at most $e(2n - e + 1) \leq 3Hn^{3/2}$. The reported optimizer comparison would then give

$$s_{m+1} \leq \frac{15}{4} s_m + 4Hn^{3/2}.$$

Since $n = 2^{m+1}$ and $2^{3/2} < 15/4 < 4$, this yields $s_m = O((15/4)^m)$ and eventually any target exponent $0 < c < \log_2(16/15)$. The strict inequality absorbs fixed constants. This is a conditional consequence. No uniform choice of $C^\dagger$, root core, and support bound is proved in the retained record.

Recall check

1. Why do 4^m pairs with 3^m norms not settle the problem? **The coordinate set creates additional ordered pairs whose outputs must be counted.**
2. What does the square-class decomposition still leave to control? **The actual signed-root support inside the base field.**
3. What is missing beyond the finite levels? **RC15/4 with fixed constants on every sufficiently large level of the actual adaptive sequence.**

STUDY NOTE 5

T04 - cubic slices and complementary units

Prepared 2026-09-14 from the record frozen through 2026-09-10. Target:

$$P_-(x,y) = (x-y)^3 - 3(x+y).$$

OPEN. The current complementary-unit construction has proved exponential input growth. A strict output exponent gap remains unknown. This target is kept separate from T01's cubic with the plus sign.

What the two papers contributed

The unit-distance work supplied the model of producing many structured relations through algebraic units. The sum-product work supplied the model of organizing units and real projections with explicit size and distortion costs. T04 tested quadratic comparisons, rich fibers, trace identities, and ultimately a fixed pair of complementary units. The enduring issue is that many relations or many inputs do not themselves bound the full output set. Sources: **S-T04**; shared foundations.

SN-T04-1 - quadratic comparison and the slice identity

PROVED, reported identities and scoped obstructions. Source: **S-T04**, quadratic-to-cubic and slice sections; **S-REG**.

The historical quadratic benchmark is $R(x,y) = (x-y)^2 + 5(x+y)$. With $u = x - y$,

$$5P_-(x,y) = 5u^3 + 3u^2 - 3R(x,y).$$

This identity still depends on u. A small set of quadratic values alone therefore does not bound the cubic image. For the specified centered box carriers, the retained results give actual injectivity for two named primes in the $5 \bmod 12$ regime and an image of size at least $|A|^2/(1+e^{-\delta d})$ in the stated $1 \bmod 12$ regime and parameter range. These are obstructions for those carriers; they are not universal cubic impossibility results. Precise carrier hypotheses remain those of **S-T04**.

The slice identity makes the missing overlaps especially clear. Put $A_u = A \cap (A-u)$, so $y \in A_u$ means $y, y+u \in A$. Then

$$P_-(A,A) = \bigcup_{u \in A-A} (u^3 - 3u - 6A_u), \qquad \sum_u |A_u| = |A|^2.$$

Each individual slice is injective in y. Cross-slice collisions must satisfy

$$6(y-y') = (u-u')(u^2 + uu' + u'^2 - 3).$$

Consequently, merely finding many rich slices or selected unit-generated translates does not give an upper bound for this union.

SN-T04-2 - fibers, local bounds, and traces: lessons that survived

PROVED, reported identities and scoped failures; OPEN broader mechanisms. Source: **S-T04**, coverage, local CRT, and trace sections; **S-REG**.

For any map on A^2, if $r(z)$ counts the pairs producing z, then

$$|f(A^2)| = \sum_{(x,y)\in A^2} \frac{1}{r(f(x,y))}.$$

This is a useful support criterion. If at most $Cn^{2-\kappa}$ ordered pairs belong to fibers smaller than n^{ε}, then $|f(A^2)| \le (C+1)n^{2-\min(\varepsilon,\kappa)}$ for $n = |A|$. By contrast, a fixed positive fraction of rich pairs can leave a quadratic tail. The recorded rich-fiber family makes this concrete: for $n = 2N$, its actual image has $4N^2 - N + 1$ values despite its deliberately rich fiber.

The exact local theorem in the specified depth ratio $\mathbf{2:3}$ is an image **lower** bound under its frozen modulus and coefficient hypotheses. It is an obstruction, not a compression theorem. The broader deep-precision BP2 statement remains open. The product-tree lower-bound argument in this record is internal; an earlier attribution to the external sum-product paper is corrected in **S-REG**, conflict MC-02.

There is also an exact norm-trace reformulation. For a finite positive rational set $H = H^{-1}$, let $A_H = \{h^2 + h^{-2} : h \in H\}$ and

$$\alpha(t) = (t - t^{-1})^3 + \sqrt{3}(t + t^{-1}).$$

The recorded identity is

$$\mathrm{Tr}_{\mathbb{Q}(\sqrt{3})/\mathbb{Q}}\big(\alpha(h/k)\overline{\alpha(hk)}\big) = 2P_-(h^2 + h^{-2}, k^2 + k^{-2}),$$

with $|A_H| = (|H| + \mathbf{1}_{1\in H})/2$. The argument bounds multiplicity of the relevant nonzero products before taking traces by 72. A bound before the trace does not bound the trace support from above. The missing trace compression remains an open condition; the identity itself is not a construction with a certified gap.

For fixed algebraic digit bases $\lambda \notin \{0, 1, -1\}$ and the specified finite graph languages of positive entropy, another recorded obstruction gives $|P_-(A_m, A_m)| \ge c(m+1)^{-2V}|A_m|^2$. Its hypotheses concern that fixed-base digit model. They should not be silently applied to every affine system with different edge multipliers.

SN-T04-3 - the complementary-unit input construction

PROVED, reported; the companion input argument inspected for exposition. Sources: **S-SACC-INPUT**, **S-SACC-AUDIT**, and **S-T04**, final frontier. No independent audit or computation was performed for these notes.

Set $\epsilon = 2 + \sqrt{3}$, $a_0 = \sqrt{3}/2$, and

$$f_+(x) = \epsilon x + a_0, \qquad f_-(x) = \epsilon^{-1}x - a_0, \qquad A_0 = \{0\}, \quad A_{m+1} = f_+(A_m) \cup f_-(A_m).$$

All inputs lie in the fixed field $\mathbb{Q}(\sqrt{3})$ with denominator at most two; cubic outputs have denominator at most eight. There are 2^m histories, but histories can coincide as scalar inputs: $+-+$ and $-+-$ already both give $-3/2$ at depth three in the companion's composition convention. Thus history count alone is not a valid input count.

The input proof extracts separated block families. Normalize $z = x/a_0$. The blocks $B_k = +(+-)^k$, for $k = 0, 1, 2$, have lengths one, three, and five and act as

$$z \longmapsto \epsilon z + 1 - k(1 - \epsilon^{-1}).$$

For a fixed number of blocks, the leading different digit dominates the remaining tail because $2/(\epsilon - 1) < 1$. The block words of total length m obey a recurrence with steps one, three, and five. Grouping them by block count loses at most a factor $m + 1$. If $\Lambda > 1$ is defined by $\Lambda^{-1} + \Lambda^{-3} + \Lambda^{-5} = 1$, the proof gives

$$|A_m| \ge \frac{\Lambda^{m-2}}{m+1}.$$

The companion audit also bounds occupation arrays and hence affine maps. With $\varphi = (1+\sqrt{5})/2$, it gives $|A_m| \le 2\varphi^4\varphi^m$. If F_m denotes the set of affine maps represented at depth m, then $|A_m| \le |F_m| \le (m+1)|A_m|$. Submultiplicativity of the map counts yields an actual input entropy

$$a_* := \lim_{m\to\infty} \frac{1}{m}\log|A_m|, \qquad \log\Lambda \le a_* \le \log\varphi.$$

This is genuine input-side progress in the supplied proof system. It does not establish an output saving.

SN-T04-4 - what pair states tell us, and what they do not

PROVED, reported transition identities; OPEN output bound. Source: **S-SACC-PAIRS**, together with **S-SACC-AUDIT**.

For a pair (X, Y), write $u = X - Y$, $s = X + Y$. Applying the same branch $\sigma \in \{1, -1\}$ to both coordinates gives

$$u' = (2+\sigma\sqrt{3})u, \qquad s' = (2+\sigma\sqrt{3})s + \sigma\sqrt{3}.$$

Applying opposite branches, with σ assigned to the first coordinate, gives

$$u' = 2u + \sigma\sqrt{3}(s+1), \qquad s' = \sigma\sqrt{3}u + 2s.$$

These affine transitions act linearly on the ten monomials in u, s of total degree at most three. This finite-dimensional description is exact, but it can contain infinitely many scalar states. Ten coordinates do not imply ten states, or a small number of values of $u^3 - 3s$.

A sufficient next result would label every ordered pair using at most $K_0(m+1)^B\kappa^m$ labels, with fixed constants and $\kappa < \Lambda^2$, in such a way that equal labels imply equal scalar outputs. It would certify every strict exponent $c < 2 - \log\kappa/\log\Lambda$. The sharper actual frontier is $\limsup_m m^{-1}\log|P_-(A_m, A_m)| < 2a_*$. Neither assertion is established. The currently certified output bound is the trivial $|A_m|^2$, so the certified positive gap is zero; the actual gap is unknown.

The companions report an exact depth-12 example with 1,030 inputs and 1,058,120 output values out of 1,060,900 ordered pairs, with maximum fiber size eight. It is finite evidence, not a uniform asymptotic conclusion. The larger depth-30 input count is not used here: the reconciliation identifies an unmatched provenance gap between that quotation and retained raw data.

Recall check

1. What does the slice identity require beyond many long slices? **Enough overlap between their scalar output sets, with every ordered pair covered.**
2. What was proved for complementary units? **Exponential growth and bounds on the actual input set, including the existence of its entropy.**
3. Why is a ten-dimensional pair representation insufficient? **Dimension does not bound the number of distinct states or scalar outputs.**

REFERENCE APPENDIX

Sources, versions, and preservation record

Study index

Version **2026-09-14**. This register identifies the exact local bytes used to prepare the study notes. Hashing establishes file identity, not mathematical correctness. The notes have no authority to supersede a research checkpoint.

Source hierarchy and availability

Current user instructions govern this documentation task. For recorded mathematical state, the applicable final checkpoint/correction chain controls, then each current-state export, then the master handoff, then reconciled repository history. Formal proof evidence remains the argument itself. A status summary can report a proved result without independently proving it.

The starting orientation used CODEX_SETUP, CURRENT_PROJECT_MAP, CODEX_MIGRATION_AUDIT, and all four tracks' SPEC and STATUS files. Those existing files and their inherited uncommitted changes were preserved. Original archive identity remains in manifest.json; separate local-migration identity remains in CODEX_SOURCE_MANIFEST.json.

Private paths below are repository-relative locators. Their files remain in ignored `local-only/`; they were not copied into these notes. A Git-only handoff therefore does not include the private references. The source register records only the references needed to understand this companion, not private manifests or administrative records.

The primary sum-product paper is **not present in the recovered source bundle used here**. Phase A's supplied bibliography identifies T. F. Bloom, W. Sawin, C. Schildkraut and D. Zhelezov, *The sum-product conjecture is false for real numbers*, arXiv:2605.28781v1, 27 May 2026. This is a transcription of that bibliography, not an independently checked citation or publication-status claim. The human remarks manuscript, frozen conceptual guide, discovery record, and separate source audit named by Phase A were not available in this bundle and were not claimed as read. The local OpenAI unit-distance manuscript **was** read at the locations identified below. No network retrieval or novelty check was performed. T01's earlier arithmetic literature survey is historical work, not part of this offline documentation task.

S-REG - reconciled correction and claim libraries

These four existing documents were consulted for the relevant mechanisms, failures, gaps, and supersessions. They are state/provenance guides, not substitutes for formal proofs. All describe the 11 September reconciliation.

- `research/MIGRATION_CONFLICTS.md` - SHA-256 `cac0563ca1a3930829e2f16cb5e64e90e314c9e061fccf2bbbc3d2f90c81f1f0`.
- `research/MECHANISM_LIBRARY.md` - SHA-256 `69bf8666425623c5199ca9c04a699548ab4b4cf8dc2b0342a447e8d2465b428e`.
- `research/FAILURE_LIBRARY.md` - SHA-256 `4fe9757355ab93b5ef50a0b2c9efa91a99cf4a1094965d4c7a3196cf766d1217`.
- `research/OPEN_LEMMAS.md` - SHA-256 `4c4395037a444c0c90410a834cb05b7b04cb07afd56b5f5e90739ab6e86ef091`.

Specific boundaries retained:

- **MC-01:** older T01 digital SPEC routes are superseded by the later scoped obstructions. The old documents remain unchanged.
- **MC-02:** the residual cubic product-tree attribution is corrected to the internal three-prime checkpoint, historically cited as §D.5, Proposition 5. The reconciliation supplies that locator; this task did not audit the underlying checkpoint or attribute its lemma to the external sum-product paper.
- **MC-03:** no separately titled later final checkpoints are located for T03 or T04. Their current states plus designated companions control; no missing checkpoint was reconstructed from a title.
- **MC-04:** the T04 depth-30 count quoted in the candidate note lacks matching retained raw evidence in the reconciliation. It is omitted from the study account; no raw data were represented as newly inspected.

- **MC-05:** the standalone T03 F25 certificate and certain earlier T04 numerical constants remain unavailable at their stated locators. The notes neither invent them nor claim to have read them.

Individually identified sources

Each SHA-256 is computed from the existing file on 14 September 2026. Where a source lacks an explicit version date, the hash identifies the version without inventing one. Section locators in the notes refer to these versions.

S-UD

`local-only/migration/research_supplement/shared/03-unit-distance-proof.pdf`

Supplied OpenAI manuscript, Planar Point Sets with Many Unit Distances. No date inferred. Theorem 1.1; Proposition 2.2; Theorem 2.3; Lemmas 2.4-2.6; Proposition 3.8 and Remark 3.9. Relevant written arguments read through PDF text extraction; page 12 also rendered and visually inspected. No independent proof audit.

SHA-256: `62b5de8cacdfbc0f13ae1940aa81659cd233e6945b936612fbeab0ab8ab3d383`

S-A

`local-only/migration/research_supplement/shared/05-Phase_A_Sum_Product_Reproduction_FINAL_FROZEN.pdf`

Final frozen Phase A reconstruction. Full extracted text read, including source map and references. Supplies the sum-product definitions, inequalities, and attribution used here; does not replace primary-paper verification.

SHA-256: `55741837f06336e072b390560dadc3ab7e087a946da69599c292c671db64f514`

S-A-LOG

`local-only/migration/research_supplement/shared/06-Phase_A_FINAL_FROZEN_Change_Log.pdf`

Companion final correction log; read in full. Controls corrected lattice factors, regulator scope, directness, and parameter choices.

SHA-256: `edfead13a44af821a9d320e6dca989179972444959bca5578b5c3984b24bded1`

S-B

`research/history-sources/PHASE_B_STATE.md`

Historical Phase B synthesis, Validated abstraction principles 1-8 and Rejected principles. Read; interpreted with the master handoff and 11 September reconciliation. Its historical next-direction paragraph is not an active assignment.

SHA-256: `ec695630f89fdc587045710f560324ee6904a8169fcf5e7e261490349414ea6b`

S-C0

`local-only/migration/research_supplement/shared/Phase_C0_Corrected.md`

Dated 3 September 2026. Theorem C0.1 and §§5-6 are the formal positive rank-one argument inspected for exposition. Dependencies were not independently recertified.

SHA-256: `8ef3670bb8903b2f851b1bba594435d8635882849dae37912baadf6c49eb85be`

S-C0-LOG

`local-only/migration/research_supplement/shared/C0_CORRECTION_LOG.md`

Correction companion, read. Controls hypotheses, real-field choice, and quantitative corrections to earlier C0 versions.

SHA-256: `5da48e6372e1ea10af8d395c30faff6e7676bc546749ffa5fc36765a7b040480`

S-MASTER

`local-only/migration/sources/00_MASTER_HANDOFF.md`

Migration snapshot dated 11 September 2026; read in full. Shared architecture in §§4-7 and novelty caution in §8. Track-specific final records outrank its summaries where applicable.

SHA-256: `b5ea32664e1a755faa4b29ad21848e11f3b16eef94b451d5b3ed5e9d18f6b250`

S-T01

`local-only/migration/sources/T01_CURRENT_STATE.md`

T01 state export, research cutoff 10 September 2026. §§4-6 support historical successes and scoped failures; §§7-8, 12-15 identify remaining gaps. Reported proof status retained; old cubic proofs were not independently audited here.

SHA-256: `2095cb7b4a5f7d848dea1356956ad36f5303e4037b13b2efe7fa32b0c08f2624`

S-T01-FINAL

`local-only/migration/sources/T01_Norm_Trace_Quadratic_Research_Checkpoint.md`

Dated 10 September 2026, before the T01 export. Separate $P_0=x^2+xy+y^2+x$ branch. §§2.1-2.3 transfer argument and §§5-7 family/rate calculations read for exposition; §8 final verdict. It cannot replace the cubic target.

SHA-256: `a9de1edb1ebc43465de7e12cd2e815e8647b4199a0c303afff2e4a8815abc0db`

S-T02

`local-only/migration/sources/T02_CURRENT_STATE.md`

T02 state export, research cutoff 10 September 2026; read. §§4-6 local progress, §§7-11 global carrier and rates, §§12-13 failures/open obligations, §19 evidence boundaries.

SHA-256: `e4ba2cbf3871d71ac45526fa7b16a0f6e468f13cce13261adafec5547830f480`

S-T02-FINAL

`local-only/migration/sources/T02_C4_REVIEW_ACCEPTANCE_AND_HANDOFF(1).md`

Dated 10 September 2026; read in full. Accepted state, Companion audit and provenance, and final handoff. Records acceptance of the C4 analytic proof with the specified failure scope; distinguishes external review from internal companion checks.

SHA-256: `8d093d4111e04dfd42d6bf13117305809beffceb44f1e8d81f16b64d4ef7c714`

S-C2-LOG

`local-only/migration/research_supplement/T02/phase_c3/C2_PRECISION_ADDENDUM.md`

Seven-point C2 precision correction companion; read. C2 formal results are reported here through the state and its correction chain, not claimed as independently audited.

SHA-256: `c747922830646d3311cb106ac2b11b7ee80e30e3a8c99696df71bfa0d378d032`

S-C3-LOG

`local-only/migration/research_supplement/T02/phase_c3/C2_C3_REVIEW_CHANGE_LOG.md`

C2/C3 review corrections supplied 8 September 2026; read. C3 universal optimization is reported from the retained source system; its full original proof was not independently audited for this task.

SHA-256: `016afcd90e64b0d22f5267e3f1f05960465baacf19cf080687bfeebd0a0dc901`

S-C4

`local-only/migration/research_supplement/T02/phase_c4/Phase_C4_Higher_Ramified_Precision.md`

C4 analytic report accepted by the 10 September handoff. §§1-4 actual local ring/count; §§5-6 signed gate and covers; §§7-8 checks and scope. Written local and rate arguments inspected. Programs were not executed.

SHA-256: `0c438ad31f11c48ff8a74bff5454bf1eda8dd7cc3b5f8b2cd05e7dd6b582a91c`

S-C4-RATE

`local-only/migration/research_supplement/T02/phase_c4/C4_SIGNED_RATE_LEDGER.md`

Dated 10 September 2026; explicitly a documentation extraction of existing rates, not a new optimization. Read in full; used for the missing quantitative support obligation.

SHA-256: `76afd88f997cdf5385dda7463654e3a164ef8044ed86cd066ef8e0fd52b4fbd8`

S-T03

`local-only/migration/sources/T03_CURRENT_STATE.md`

Controlling T03 current state, research cutoff 10 September 2026; read. §3 adaptive definitions; §§4.1-4.7 exact identities and finite results; §§5.2-5.10 historical families; §6 scoped failures; §7 RC15/4; §§13-15 final frontier and limitations.

SHA-256: `8048b4d9d3121757a28abe90e227fa5fba84cb4eb172bef8b6aa7186b3792850`

S-T03-ZIP

`local-only/migration/research_supplement/T03/T03_RESEARCH_RECORD_20260910.zip`

Frozen supporting archive. Member inventory inspected before selective extraction of the five Markdown reports listed below. Those reports were read in full. No Python/C++ program, JSON computation, or pickle object was executed or deserialized.

SHA-256: `5c51114ec0fa2663b586665db30bc6648f1e3a1beaae2e578099b9abc6e85430`

S-T04

`local-only/migration/sources/T04_CURRENT_STATE.md`

Controlling T04 state, research cutoff 10 September 2026; read. §§4.3-4.8 supply local, transfer, trace, coverage and digit claims; §4.9 and §5.10 give current input progress; §7 and §§13-15 distinguish active from historical open endpoints.

SHA-256: `17d5bb792fc4a9903bac337e0b96d283bad551796ddccaa3b8965816b8ac59b5`

S-SACC-INPUT

`local-only/migration/research_supplement/T04/sacc_candidate_input.md`

Designated final affine companion. Read in full: fixed system, separated-block lower bound, occupation upper bound and finite evidence. Large quoted numerical data are subject to MC-04 below.

SHA-256: `e98ebe18d94698a1d79f8f9157f61e455723c31cc18dc2f803189642e2daae28`

S-SACC-AUDIT

`local-only/migration/research_supplement/T04/sacc_input_audit.md`

Designated input-proof audit companion, read in full. §§1-4 give graph convention, proof checks and entropy limit; §5 explicitly leaves output compression open. This is an inherited internal audit, not an audit performed for these notes.

SHA-256: `938ea55d596119fe269d06b80864864c97606eb4f7f128256b2b10905a0fc10e`

S-SACC-PAIRS

`local-only/migration/research_supplement/T04/sacc_pair_states.md`

Designated pair-state companion, read in full. Exact transitions, finite-dimensional closure, inherited finite counts and missing output certificate. No computations rerun.

SHA-256: `dd3af5e907e7a8feff5de9fbaee3bef4a6407bb97849ac83f13c6f1db7539355`

Selected T03 archive members

These member identities refine **S-T03-ZIP**. Only their Markdown text was extracted to ignored scratch storage and read.

- `output/TCRC_Finite_Covers_m1_m2.md` - SHA-256 `40d6446b9b4f8bc4bb2ee2c6cc3491b7f6f3beca15ddddab3979f13671a4280f`.
- `output/TCRC_Finite_Covers_m3_m4.md` - SHA-256 `b828d2ee5a2a101dae285627f42bcfa7bec23d5b564054565c6c3e2d319ee65c`.
- `output/TCRC_History_Parameters.md` - SHA-256 `e39299e1c43335bf26e1c28549b3f6eab20edf69fb4ea0aef91cfd7953bf825a`.
- `output/TCRC_Root_Algebra_Pair_Maps.md` - SHA-256 `483a232f4344c9c354e02fd21e8f7ab78cc44557ff48ad02306600c079ac05f0`.
- `output/TCRC_Triangle_Closure.md` - SHA-256 `f6586783e413a57cebc4651cd32a48fe5a0e4a50615ef86af5ef6c23b08f6537`.

Claim-to-source map

The SN identifiers are documentary study items dated 14 September 2026. Quantifiers, polynomial, hypotheses, dependencies, and gaps appear in the corresponding note; this table supplies the principal proof/status locators. No SN item asserts a new construction or independently audited theorem.

Study items	Principal locations and evidence
SN-S1-S3	S-UD named theorem/lemmas and Proposition 3.8; S-A §§3-8, 10-13 and S-A-LOG; supplied arguments inspected for exposition
SN-S4	S-B Validated abstraction principles 1-6; S-MASTER §§5-6; reported abstraction with displayed counting inequalities
SN-S5	S-Co Theorem Co.1, §§5-6 and S-Co-LOG; written proof inspected, dependencies not independently audited
SN-T01-1-3	S-T01 §§4.1-4.5 and §6, its named earlier proof locators, S-REG MC-01; source-reported proofs/failures
SN-T01-4	S-T01-FINAL §§2.1-2.3, 5-8; transfer and rate arguments inspected; premises remain open
SN-T02-1-2	S-T02 §§4-6 and C2/C3 correction chain; S-C4 §§1-4; reported earlier proofs and inspected C4 local argument
SN-T02-3-4	S-C4 §§5-6, S-C4-RATE and S-T02-FINAL; scoped certificate failure and open replacement obligation
SN-T03-1-2	S-T03 §§5-6; reported identities, scoped obstructions, and finite exact certificates
SN-T03-3-4	S-T03 §§3-4 and five S-T03-ZIP reports; exact algebra and finite certificates read, programs not rerun
SN-T03-5	S-T03 §7 RC15/4, §4.7 conditional recurrence and §§13-14; open premises and their conditional payoff
SN-T04-1-2	S-T04 §§4.3-4.8 and §5; source-reported historical identities/obstructions; S-REG MC-02
SN-T04-3-4	S-SACC-INPUT, S-SACC-AUDIT §§1-5, S-SACC-PAIRS; input and pair arguments inspected; scalar output gap open

Work performed and limits

This task added only the seven Markdown files in this dated study folder. Temporary PDF text, one page render, and a preservation baseline were stored under ignored `scratch/study-notes-2026-09-14/`. No existing mathematical source, STATUS, SPEC, manifest, archive, program, or inherited change was edited. No imported program was executed, no pickle was deserialized, and no new mathematical experiment, proof search, external review, or literature search was undertaken. The final checks below concern documentation and file identity.

Validation on 14 September 2026:

- `python3 scripts/check_repository.py` - passed; verified 39 archived sources and local links in 40 Markdown files covered by that checker.
- `python3 scripts/check_codex_setup.py` - passed; verified 42 local references, original import identities, 43 T03 manifest members, duplicate states and C4 hashes, without executing research code.
- `git diff --check` - passed for the inherited tracked diffs.
- A separate check of the seven new Markdown files found no broken relative links, missing source IDs, trailing whitespace, or unbalanced display delimiters.
- SHA-256 comparison against the task's starting snapshot found all 90 pre-existing nonignored repository files unchanged. The only nonignored additions were these seven notes. Tracked diffs therefore remain inherited.

These checks do not prove any research target. The files remain uncommitted.